

Project 1: Knowledge Retrieval (part - 2 Implement Monte Carlo Tree Search (MCTS) Algorithm)

Introduction

This part of project includes Monte Carlo Tree Search algorithm(MCTS) to retrieve a task tree based on the user's goal, optimized through simulations to determine the best steps based on success rates.

Features

- Load and process a FOON graph from a text file.
- Specify goal objects (e.g., dishes or foods), their states, ingredients, and containers.
- Performs Monte Carlo Tree Search algorithm(MCTS) through FOON nodes to find the best sequence of steps required to achieve the goal.
- Saves the generated task tree to an output file.

Usage

To run the program, follow these steps:

1. **Run the main script:**

```
python test_script.py
```

Follow the prompts:

1. Enter the path to the FOON file - i.e, FOON.txt
2. Enter the goal object's name - for example: ice
3. Enter the goal object's state(s) (comma-separated if multiple).
4. Enter the ingredients (comma-separated if multiple).
5. Enter the name of the container required for the goal object.

Once the process is completed, the system will search for the goal object and its matching sequence in the FOON graph. It generates an output task tree saved in a .txt file containing the recipe steps.

Output Screenshot:

For goal node i.e, ice

```
PS E:\Vignesh Ambeeru\Vignesh Ambeeru> python test_script.py
Enter the path to the FOON file (e.g., 'FOON.txt'): FOON.txt
-- universal foos saved to FOON.pkl
-- total functional unit: 2376
Enter the goal object name: ice
Enter the goal object state (comma-separated if multiple): crushed, frozen
Enter the goal ingredients (comma-separated if multiple):
FOON data loaded. Functional units: 2376, Object nodes: 4305
Enter the container: drinking glass
writing generated task tree to output mcts_ice.txt
PS E:\Vignesh Ambeeru\Vignesh Ambeeru>
```